



4. **Select diagonal size:** Round to the nearest common screen size (e.g., 55", 65", 75").
- Example:** If your sofa is 10 ft (120 in) from the screen:
- **For 4K:** 120 in ÷ 1.6 = 75" diagonal.
 - **For 1080p:** 120 in ÷ 2 = 60" diagonal (within the 1.5–2.5 range).
- Consider ceiling height, furniture placement, and sight-lines so the centre of the screen aligns roughly with eye level when seated.

Resolution: From 1080p to 8K

The choice of resolution determines perceived sharpness and future-proofing. Full HD (1920 × 1080) delivers about 2 million pixels, while Ultra HD or 4K (3840 × 2160) boasts over 8 million pixels – quadrupling detail. 8K UHD (7680 × 4320) pushes pixel count to over 33 million, but currently 8K content remains scarce and costs are extremely prohibitive for mass adoption.

- **1080p (2 Mpx):** Suitable for 40"–55" screens if viewing distance exceeds 2.5 × diagonal.
- **4K UHD (8 Mpx):** Ideal for 55"–75" displays at 1.5–1.6 × diagonal, and future-proof for streaming content.
- **8K UHD (33 Mpx):** Best reserved for screens ≥75" if sitting within 1.2 × diagonal; otherwise you won't benefit.

Why 4× Pixels Matter: A 4K display packs four times more pixels than Full HD, doubling linear pixel density for sharper

fine detail. This allows closer seating without seeing the “screen door” effect.

Practical Tip: If you primarily watch 4K streaming or plan to upgrade to next-gen consoles, opt for 4K. Skip 8K unless you crave the latest tech and have generous amounts of indoor space.

Panels: LCD, OLED, QLED, and MiniLED

LCD vs. OLED

- **LCD (LED-backlit):** Uses a backlight behind an LCD panel. Brightness excels in sunny rooms, but contrast and blacks may suffer.
- **OLED:** Each pixel emits its own light, delivering true blacks and infinite contrast – ideal for dark-room viewing.

QLED and MiniLED

- **QLED (Quantum-dot LCD):** Enhances colour volume via quantum-dot layer, boosting brightness and vibrancy.
- **MiniLED:** Shrinks backlight zones for finer local dimming and better HDR contrast without burn-in risk.

Example Models:

- **LG C5 OLED:** Praised for deep blacks and 144Hz refresh rate – strong for gaming and cinema.
- **Hisense U7N QLED:** A budget pick delivering VRR and Dolby Vision at an affordable price.
- **TCL QM8K MiniLED:** Offers 3,800 dimming zones and 4K@144Hz gaming support for ₹1,59,900 (65").

Recommendation: In bright living rooms, choose high-brightness QLED or MiniLED. For dedicated home theaters, prefer OLED for its blacks and contrast.

Picture quality features

High Dynamic Range (HDR)

HDR expands contrast and color gamut. Key formats include HDR10 (base), HDR10+ (dynamic metadata), and Dolby Vision (industry-leading dynamic metadata support). Dolby Vision often yields the richest image, but HDR10+ is gaining traction on Samsung and Amazon Prime.

Feature	LED	OLED	QLED	Mini LED
Colour Volume	●●○	●●○	●●●	●●●
Contrast Ratio	●○○	●●●	●○○	●●○
Viewing Angle	●○○	●●●	●●○	●●●
Brightness	●●○	●●○	●●○	●●●
Ideal For	Everyday streaming in bright or dark rooms	Dark rooms, nighttime viewing, and gaming	Big-screen home theatre setups in bright or dark rooms	Bright rooms and daytime viewing